TransLink Transit Intelligence Pipeline

NFO 4190- PROJECT INTEGRATION 1 | MIDTERM REPORT

Group 9

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1.0 Project Title and Team Info

Project Title: TransLink Transit Intelligence Pipeline: System Design and Architecture for a Data Engineering and Machine Learning Solution for Public Transit Schedule Analysis and Commuter Departure Insights in Metro Vancouver

Name	Student ID	Role
Inderpreet Kaur	100383066	Technical Design Lead, system architecture, data pipeline design, database schema, ML model design, API design
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2.0 Introduction and Objectives

Metro Vancouver's transit network, run by TransLink, moves hundreds of thousands of commuters a day across buses, SkyTrain, SeaBus, and RapidBus. Apps like Google Maps and TransLink's own Trip Planner are great at answering "when's the next bus?" but none of them answer a different question: *when should I travel?*

A commuter going from Surrey to Vancouver on a Sunday afternoon has no way of knowing if noon is the busiest window on their route, or if leaving two hours earlier would mean a quieter ride. Answering that means crunching hundreds of thousands of scheduled trips across the network, something no public tool currently does.

TransLink does publish its full schedule weekly as open GTFS data, but it's not usable by anyone without programming skills, it's a set of large, structured text files that need real technical know-how to parse and interpret.

The TransLink Transit Intelligence Pipeline closes that gap: an end-to-end pipeline that automatically pulls TransLink's weekly GTFS data, applies machine learning to spot service patterns, and surfaces it all through a simple, commuter-friendly dashboard.

2.1 Refined Scope as of Week 7

The scope hasn't changed much since the proposal, just three clarifications based on professor feedback:

- K-Means clustering is now an optional stretch feature, not core. The real value, schedule-frequency aggregation by route, stop, day, time, and direction, works fine without it.
- All documents and planned UI copy now consistently use "scheduled service frequency" and "service intensity," since the system can't actually confirm real passenger demand or crowding, only what TransLink has scheduled.
- User testing is now a formal success metric: we'll confirm the plain-language departure recommendation makes sense to non-technical users with no explanation needed.

2.2 Target Users

- Metro Vancouver daily commuters seeking smarter departure planning
- Transit researchers and urban planners studying network-wide service patterns
- Open data enthusiasts interested in civic technology applications

2.3 Assumptions

- TransLink continues publishing GTFS data weekly.
- GTFS file structure remains largely unchanged.
- Users have internet access and modern web browsers.
- Scheduled service frequency can reasonably serve as a proxy for transit service intensity.
- This strengthens the design documentation.

3.0 Requirements Summary

The following requirements reflect the refined scope as of Week 7. Three changes have been made since the original proposal: K-Means clustering reclassified as optional, language standardized around scheduled service frequency, and user testing added as a formal success metric.

3.1 Functional Requirements

#	Requirement	Priority
FR-01	The system shall automatically download the TransLink GTFS Static ZIP file from the published URL every Friday	Core
FR-02	The system shall parse and load all core GTFS files (routes, trips, stop_times, calendar, stops) into PostgreSQL without errors	Core
FR-03	The system shall check feed_info.txt after each download and skip re-import if the feed dates have not changed	Core
FR-04	The system shall allow users to search for a route by number or name	Core
FR-05	The system shall allow users to search for a stop by name and view its scheduled service patterns	Core
FR-06	The system shall display scheduled service frequency by route, day type (weekday/Saturday/Sunday), time window, and direction	Core
FR-07	The system shall generate a plain-language departure suggestion for every Route Analyser result	Core
FR-08	The system shall display a data refresh status indicator showing when the GTFS feed was last processed and its validity dates	Core
FR-09	The system shall support filtering by route, stop, day of week, time range, transit mode, and direction	Core
FR-10	The system shall display a disclaimer on every dashboard page clarifying that results reflect scheduled service frequency only, not real-time ridership or crowding	Core
FR-11	The system shall display service intensity charts visualizing scheduled frequency patterns for a selected route and day type	Core
FR-12	The system shall support filtering trips and stops by wheelchair accessibility using the wheelchair_accessible and wheelchair_boarding columns already present in the TransLink GTFS dataset	Stretch
FR-13	The system shall allow users to compare scheduled service frequency across two routes or two departure windows	Stretch

FR-14	The system shall apply K-Means clustering to group routes by service behaviour and display results on the Network Patterns dashboard page	Stretch

3.2 Non-Functional Requirements

#	Requirement	Category
NFR-01	Core dashboard pages shall load within 3 seconds on a standard broadband connection	Performance
NFR-02	The stop_times.txt file (87.78 MB) shall be loaded using Pandas column-selective loading to reduce memory usage	Performance
NFR-03	All UI copy, plain-language summaries, and design documents shall consistently use "scheduled service frequency" or "service intensity", never "busy," "crowded," or "demand" without qualification	Usability
NFR-04	The plain-language departure recommendation shall be correctly interpreted by at least 6 out of 8 test participants without any explanation from the development team	Usability
NFR-05	The system shall remain functional if TransLink does not publish an updated feed on a given Friday, serving the most recently loaded dataset until a new feed is available	Reliability
NFR-06	All design artefacts and source files shall be version-controlled in a shared GitHub repository using consistent naming conventions agreed upon in Week 2	Maintainability

3.3 Project Success Criteria

Success Metric	Target
GTFS feed imports successfully	100% of weekly imports
Dashboard page load time	Under 3 seconds
Route search response time	Under 2 seconds
User testing comprehension	6/8 users correctly interpret recommendations
Weekly refresh automation	Runs without manual intervention

4.0 Literature Review and Research

This section covers five sources: TransLink's own GTFS platform, two academic studies on transit/movement data, a popular transit app, and a recent BRT scheduling simulation.

[1] TransLink GTFS Open Data and Trip Planner
TransLink's GTFS feed contains 63,250 trips across 242 routes, and its Trip Planner gives real-time next-departure info [1]. But the two don't connect for a regular commuter, the raw data needs programming skills to touch, and the Trip Planner only shows the next few buses, never whether your chosen time is busy or quiet. This project bridges that gap by turning the open GTFS data into plain-language scheduled-frequency insights, no technical background required.

[2] Antrim and Barbeau (2013) - The Many Uses of GTFS Data
Antrim and Barbeau showed GTFS data supports far more than trip planning, accessibility analysis, service mapping, multimodal planning, visualization [2]. Their work was aimed at developers and researchers, though, not everyday commuters. This project extends that idea into an end-to-end pipeline that turns GTFS data into guidance non-technical commuters can actually use.

[3] Roth et al. (2011) - Structure of Urban Movements
Roth, Kang, Batty, and Barthélemy studied London Underground Oyster card data and found transit activity is highly uneven, a small set of routes and time windows carry a disproportionate share of movement [3]. That supports this project's core assumption: service intensity varies a lot by route and time, and surfacing that has real value. TransLink doesn't release ridership data publicly, though, so this system uses scheduled service frequency as a proxy, a standard, accepted approach in transit research, consistent with Roth et al.'s methodology [3].

[4] Transit App
Transit integrates GTFS and real-time feeds for live vehicle tracking, routing, and arrival predictions [4]. Google Maps Transit does something similar. Both are great for "when's the next bus," but neither analyzes scheduled patterns to help users pick a better time to travel, exactly the gap the Route Analyser and plain-language summary are built to fill.

[5] Altarazi and Janho (2026) - Optimizing Bus Rapid Transit Systems
Altarazi and Janho built a discrete-event simulation to optimize BRT scheduling in Amman, Jordan, balancing wait times against fleet utilization [5]. They found inconsistent headways can increase peak-hour wait times by 15–30%, and that scaling fleet size down off-peak is the most effective way to control costs without hurting service. This backs up a core assumption here too, frequency really does vary meaningfully between peak and off-peak,

just approached from the commuter side rather than the operator side, using GTFS data TransLink already publishes weekly.

5.0 Functional Analysis and Design Considerations

5.1 Comparison with Existing Tools

TransLink Trip Planner, Google Maps Transit, and the Transit App were each evaluated against the features planned for this system.

Feature	TransLink Trip Planner	Google Maps Transit	Transit App	This System
Real-time next departure	Yes	Yes	Yes	No (by design)
Live vehicle tracking	Yes	Yes	Yes	No (by design)
Scheduled frequency by time window	No	No	No	Yes
Peak vs off-peak service analysis	No	No	No	Yes
Plain-language departure suggestion	No	No	No	Yes
Filter by day type (weekday/Saturday/Sunday)	No	Limited	No	Yes
Filter by direction	No	No	No	Yes
Stop-level service pattern analysis	No	No	No	Yes
Accessibility-aware trip filtering	No	Limited	No	Yes (stretch)
Data freshness indicator	No	No	No	Yes
Network-wide service pattern overview	No	No	No	Yes

5.2 Analysis of the Gap

All three existing tools answer "when's the next bus?" well. None answer "when should I travel?", that's the whole basis for this project.

TransLink's Trip Planner is the most feature-rich for Metro Vancouver specifically, real-time arrivals, alerts, step-by-step planning, but gives zero insight into whether a chosen time is peak or off-peak, and no historical pattern analysis. Google Maps Transit is navigation-focused: it tells you which route to take, not when conditions are better. Transit App adds live vehicle positions and countdowns but, again, no frequency analysis or timing guidance.

This isn't a flaw in those tools, they're well-built for what they're meant to do. It's a deliberate scope gap: they're built for commuters who've already picked a departure time. This system is for commuters with flexibility who want to know which window is busiest or quietest, a question none of the existing tools answer.

5.3 Feature Justification

Route Analyser & Plain-Language Summary: A conversation with a regular TransLink commuter confirmed that people travelling with strollers, mobility aids, or elderly family members actively look for quieter windows, not just the next bus, something no existing tool supports. Altarazi and Janho's finding that headway inconsistency swings wait times by 15–30% in a purpose-built BRT system [5] suggests the same dynamic likely applies here. Surfacing that in plain language is the core value of this system.

GTFS Import Pipeline & Weekly Refresh: The Week 1 proof-of-concept confirmed TransLink's GTFS data loads cleanly, 242 routes, 63,250 weekly trips, 87.78 MB of stop-level timing data [1]. Antrim and Barbeau confirmed GTFS is well-suited to this kind of analysis [2]. Since TransLink refreshes the feed every Friday, automated weekly ingestion is necessary, otherwise the system's data goes stale within days.

Direction Filtering: The Week 4/5 file analysis found TransLink's GTFS includes a non-standard directions.txt file with compass direction per route. That enables direction-aware filtering, a feature none of the three existing tools offer, and one that matters for routes where inbound/outbound patterns differ.

Accessibility-Aware Trip Filtering (Stretch): The wheelchair_accessible and wheelchair_boarding columns are already populated in the dataset, so no extra data source is needed. Given the stakeholder feedback above, this is a natural stretch feature, just not required for the core system to work.

Data Freshness Indicator: feed_info.txt gives each feed's validity dates. Showing this on every page is both a transparency move and a practical one, without it, commuters can't tell if they're looking at current or stale data.

5.4 Preliminary Database Design

Table	Purpose
routes	Stores route information including route ID, route name, transit mode, and route descriptions.
trips	Stores individual scheduled trips associated with routes and service calendars.
stop_times	Stores scheduled arrival and departure times for each stop on every trip.
stops	Stores stop names, locations, wheelchair accessibility information, and stop metadata.
calendar	Defines service availability for weekdays, Saturdays, Sundays, and holiday schedules.
feed info	Stores GTFS feed publication dates and validity periods used for refresh tracking.
directions	Stores route direction information used for inbound/outbound service analysis.

These tables represent the core GTFS entities that will be imported into PostgreSQL and used for service frequency aggregation, route analysis, dashboard visualizations, and API queries. Additional derived tables may be created during implementation to improve query performance and support analytics features.

6.0 Response to Peer Review #1

Team 8 (Jimoh Samira and Ify Ijeh) reviewed our Progress Report 1 and rated it highly across all five evaluation criteria, with scores of 5/5 for Clarity of Scope, Actionable Next Steps, and Overall Organization, and 4/5 for Relevance of Research and Identification of Risks. The reviewers noted that the project demonstrated strong technical research and validation, and that the scope refinement prioritizing core commuter value over machine learning complexity reflected sound project management.

Two specific, actionable suggestions were raised, and both have been addressed below.

Feedback 1: Include a system architecture diagram earlier in the report

Team 8 suggested that a simple system architecture diagram would help readers understand how the data pipeline, database, API, and dashboard interact, rather than waiting for this to appear only in future deliverables.

Action taken: A simplified four-layer architecture diagram has been added below, ahead of the formal, detailed architecture diagram scheduled for Weeks 7–9 per the implementation roadmap. This early version is intentionally high-level, showing the four major system layers and their primary responsibilities.

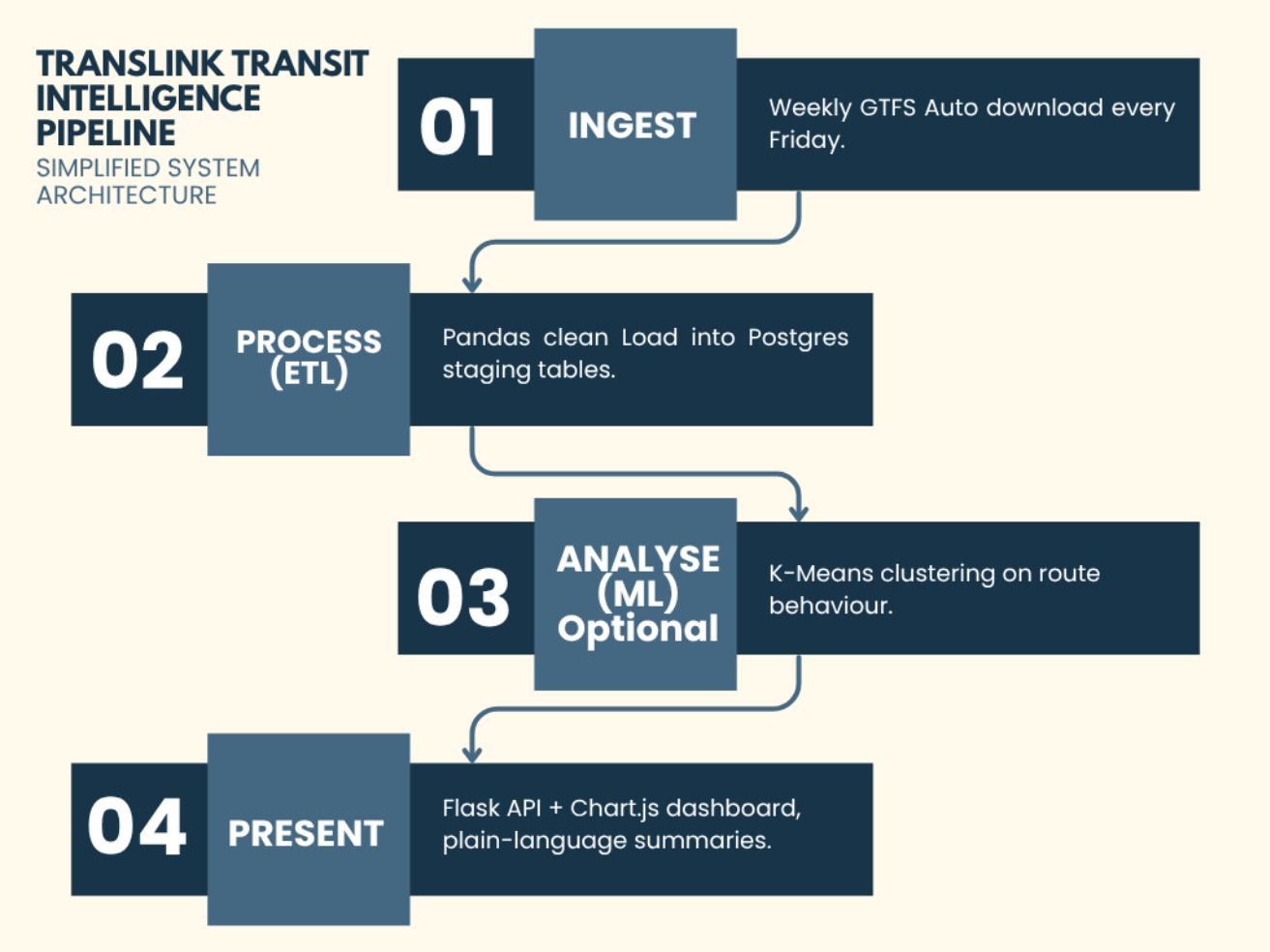


Figure 1: Simplified System Architecture

This diagram will be expanded into a full system architecture diagram, including data flow direction, database schema references, and specific API endpoints, as part of the Week 7–9 detailed design deliverables. The architecture follows a four-layer design. The Data Ingestion Layer retrieves and validates GTFS feeds from TransLink. The Data Processing Layer parses and transforms raw files into structured datasets. The Storage Layer uses PostgreSQL for persistent storage and querying. The Presentation Layer exposes data

through APIs and visual dashboards that allow commuters to explore service intensity patterns.

Feedback 2: Strengthen the risk section with user adoption and usability risks, not only technical risks

Team 8 noted that the existing risk table in Progress Report 1 focused primarily on technical challenges (GTFS format changes, K-Means clustering reliability, scope size, team coordination) and recommended adding risks related to user adoption and usability.

Action taken: The following risk has been added to the project's risk register:

Risk	Description	Impact	Mitigation Strategy
Low user adoption due to unfamiliar terminology or unclear value proposition	Commuters may not understand what "scheduled service frequency" means or may not see why this tool is more useful than apps they already use for real-time arrivals	Could result in low engagement even if the system functions correctly, undermining the project's core goal	The plain-language summary feature is designed specifically to translate technical schedule data into everyday language. The Week 7+ user testing plan (8 participants, 6/8 pass threshold) directly tests whether the value proposition is understood without explanation. Onboarding copy will explicitly contrast this tool with real-time apps to clarify its distinct purpose.

This complements the existing technical risk register from Progress Report 1 and ensures both technical feasibility and user-facing adoption concerns are tracked going into the implementation phase.

7.0 References

[1] TransLink, "GTFS Static Data," South Coast British Columbia Transportation Authority, 2026. [Online]. Available: https://www.translink.ca/about-us/doing-business-with-translink/app-developer-resources/gtfs/gtfs-data

[2] A. Antrim and S. J. Barbeau, "The many uses of GTFS data, opening the door to transit and multimodal applications," University of South Florida, 2013. [Online]. Available: https://www.researchgate.net/publication/268097444

[3] C. Roth, S. M. Kang, M. Batty, and M. Barthélemy, "Structure of urban movements: Polycentric activity and entangled hierarchical flows," *PLOS ONE*, vol. 6, no. 1, e15923, Jan. 2011. doi: 10.1371/journal.pone.0015923

[4] Transit Inc., "Transit, the best app for buses and trains," 2026. [Online]. Available: https://transitapp.com

[5] S. Altarazi and N. Janho, "Optimizing bus rapid transit systems: A simulation approach to scheduling," *Journal of Advanced Transportation*, vol. 2026, 2026. doi: 10.1155/atr/4048179